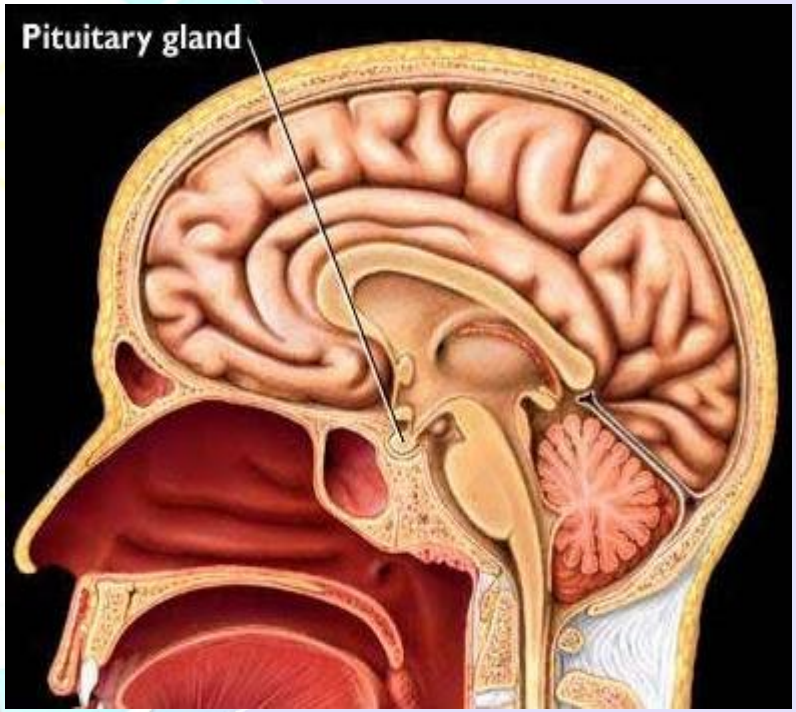
GROSS AND DEVELOPMENT OF PITUITARY

LEARNING OBJECTIVES.

- At the end of the lecture, student should be able to:
- Know the gross anatomy of pituitary gland.
- Know the development of pituitary gland.
- Know the congenital anomalies related to pituitary gland.

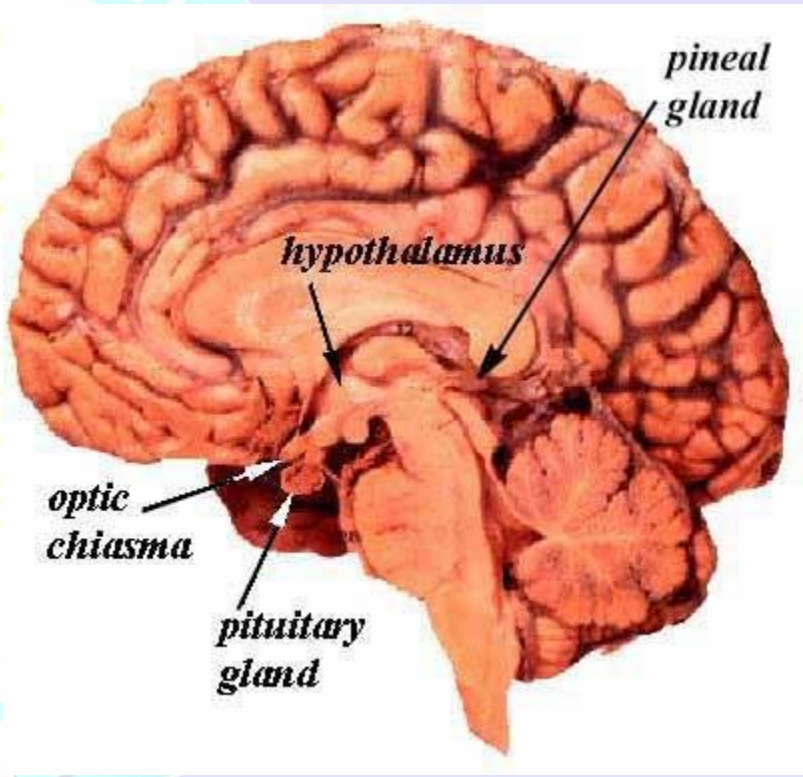
PITUITARY GLAND

- It is an endocrine gland about the size of a pea and weighing 0.5 g (0.02 oz.), in humans.
- It is a protrusion off the bottom of the hypothalamus at the base of the brain, and rests in a small, bony cavity (sella turcica) covered by a dural fold (diaphragma sellae)



PITUITARY GLAND

- The pituitary is functionally connected to the hypothalamus by the median eminence via a small tube called the infundibular stem.

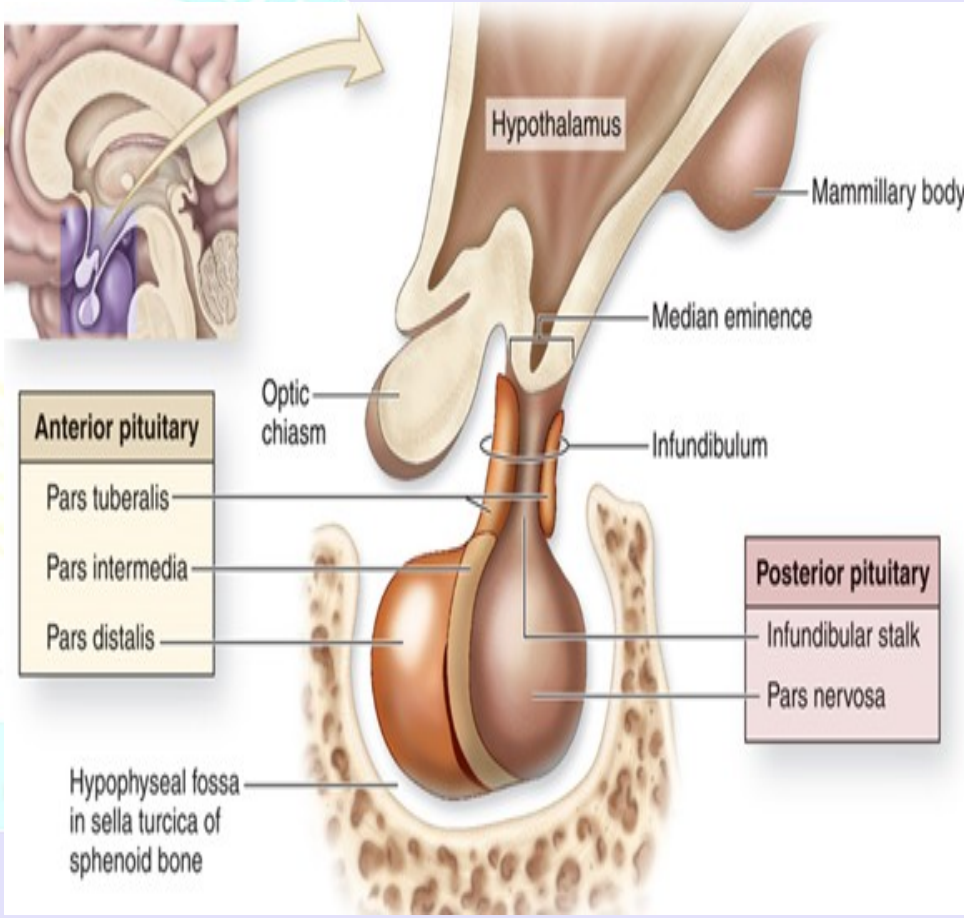


ANATOMY OF PITUITARY GLAND

- Pituitary gland lies in the base of the skull in a portion of sphenoid bone
- It consist of two lobes:

✓ Anterior lobe (adenohypophysis)

✓ Posterior lobe (neurohypophysis)



PITUITARY GLAND AND ITS RELATION TO THE HYPOTHALAMUS

PITUITARY GLAND: TWO DISTINCT PARTS

1. The Anterior (Adenohypophysis)
2. The Posterior: (Neurohypophysis) and
3. Pars intermedia

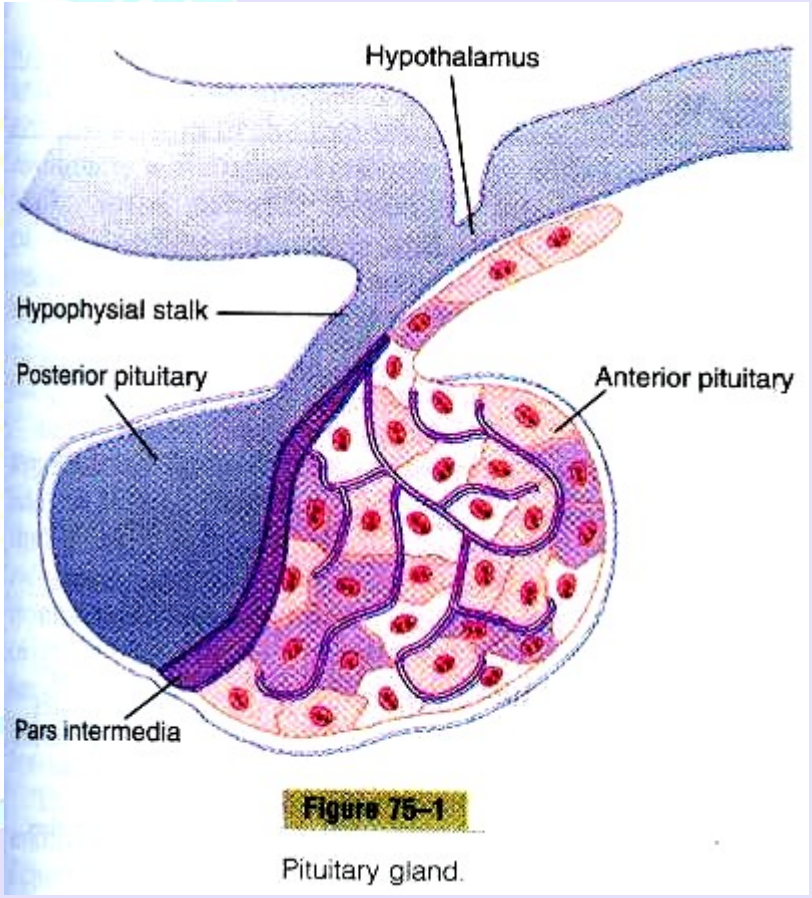
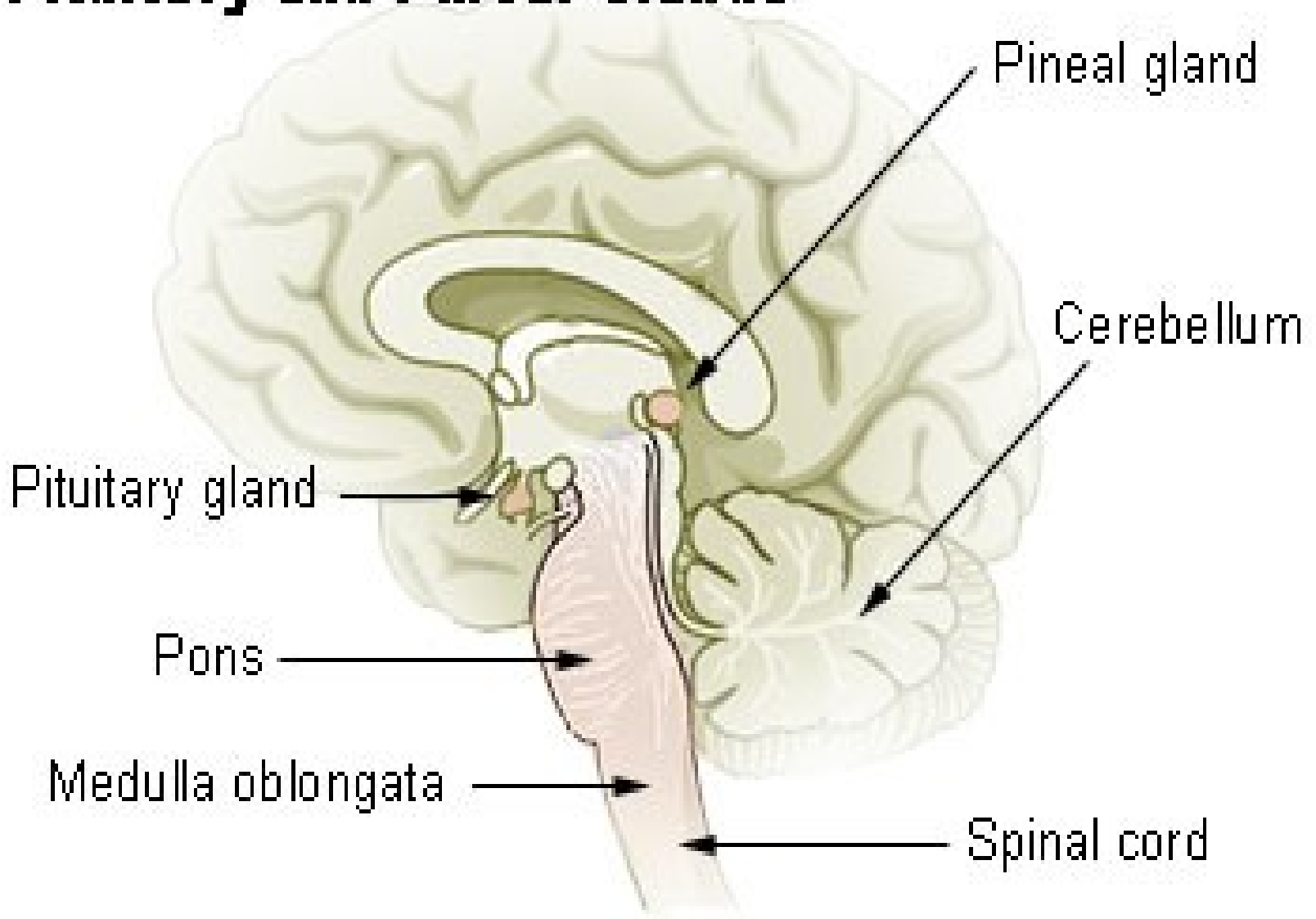


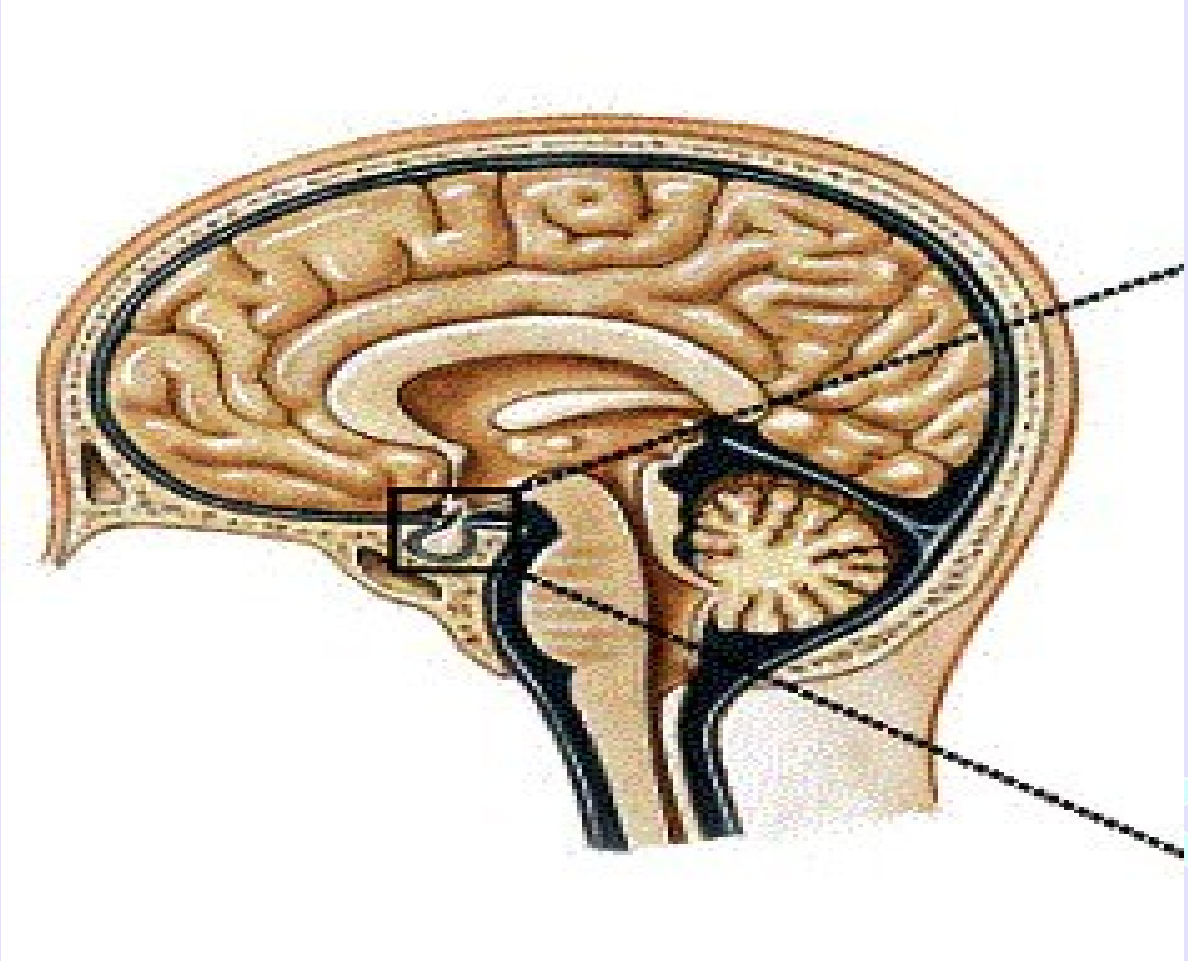
Figure 75–1

Pituitary gland.

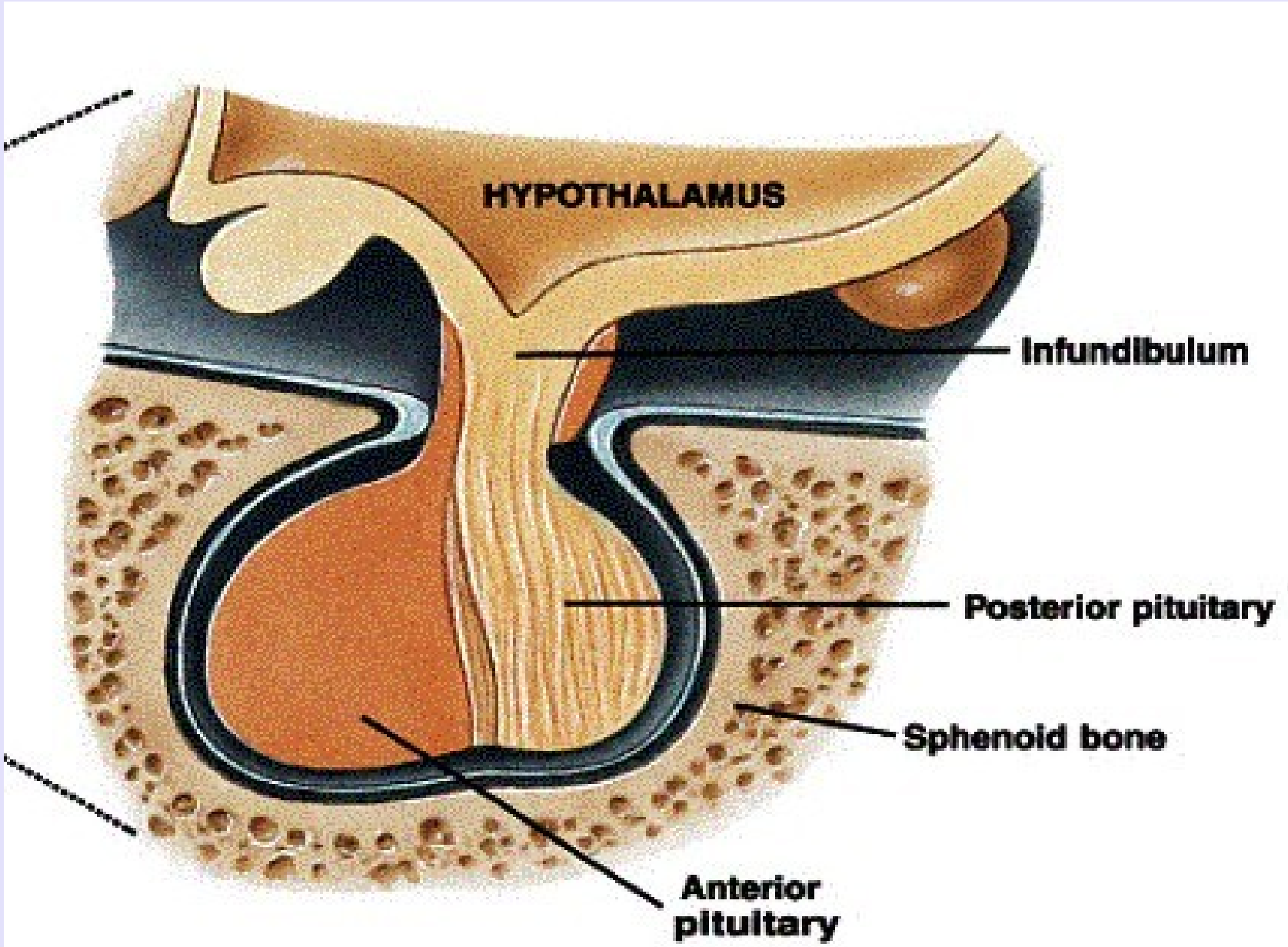
Pituitary and Pineal Glands



LOCATION OF THE PITUITARY



SELLA TURCICA OF THE SPHLENOID BONE


HYPOTHALAMUS
Infundibulum
Posterior pituitary
Sphenoid bone
Anterior pituitary

PITUTARY GLAND DEVELOPMENT

- Is Ectodermal in origin
- Develops from two sources

  a). Evagination from ectodermal roof - stomodeum or Rathke's spouch.

  b). A down growth from neurectoderm- Neurohyphysis

- Their double origin give rise to two different types of Tissues

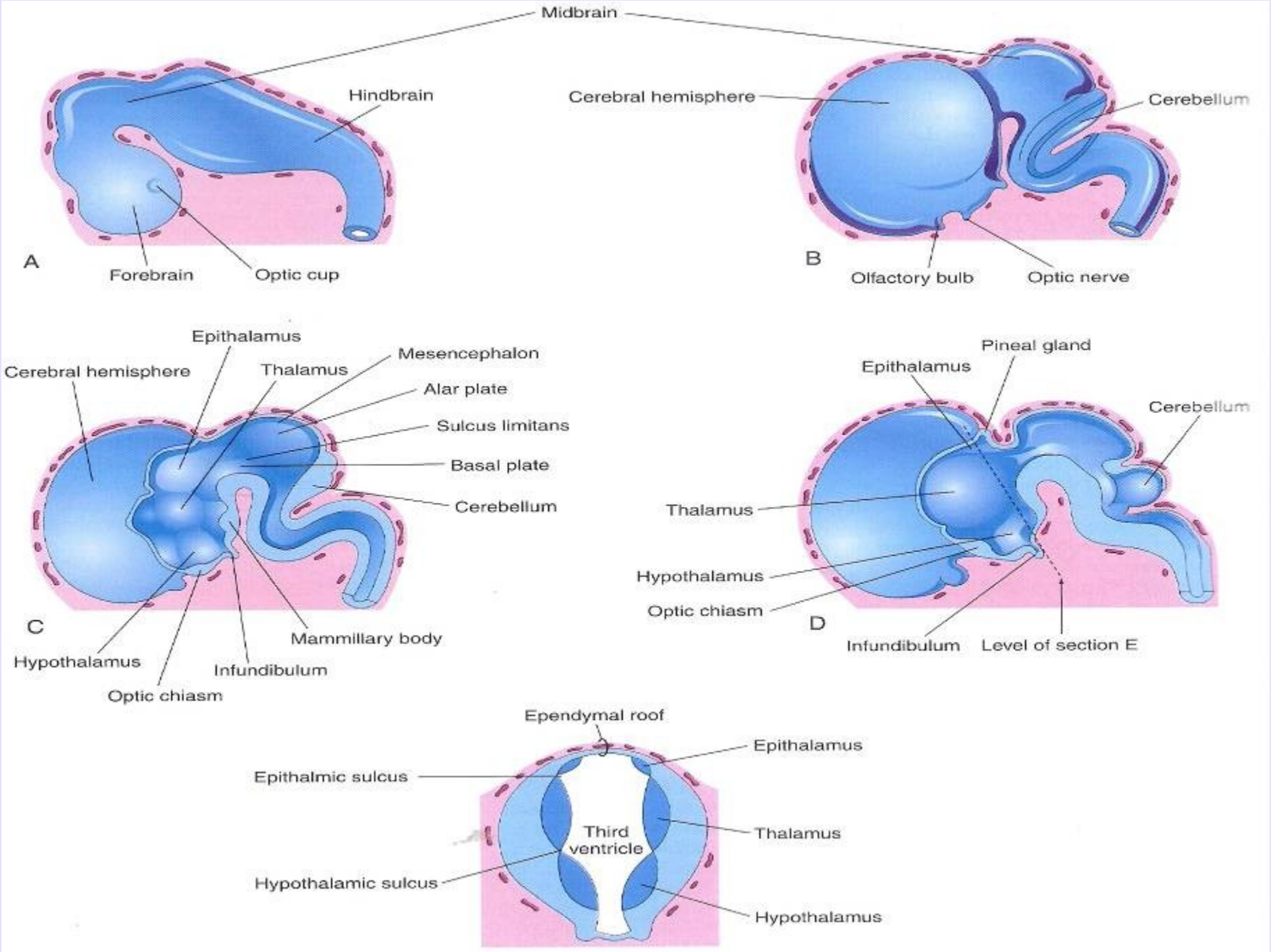
PITUITARY GLAND DEVELOPMENT

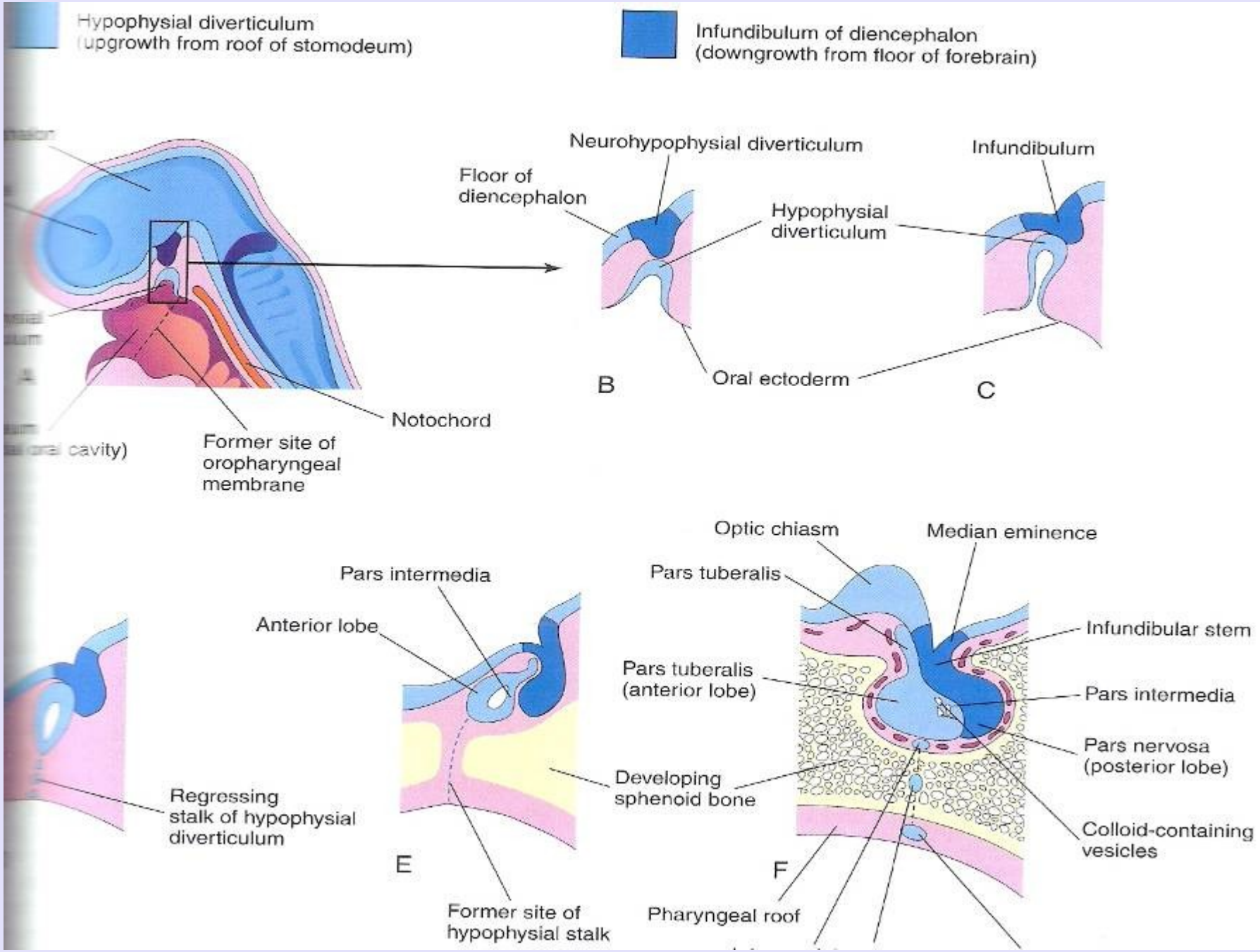
ADENOHYPOPHYSIS;

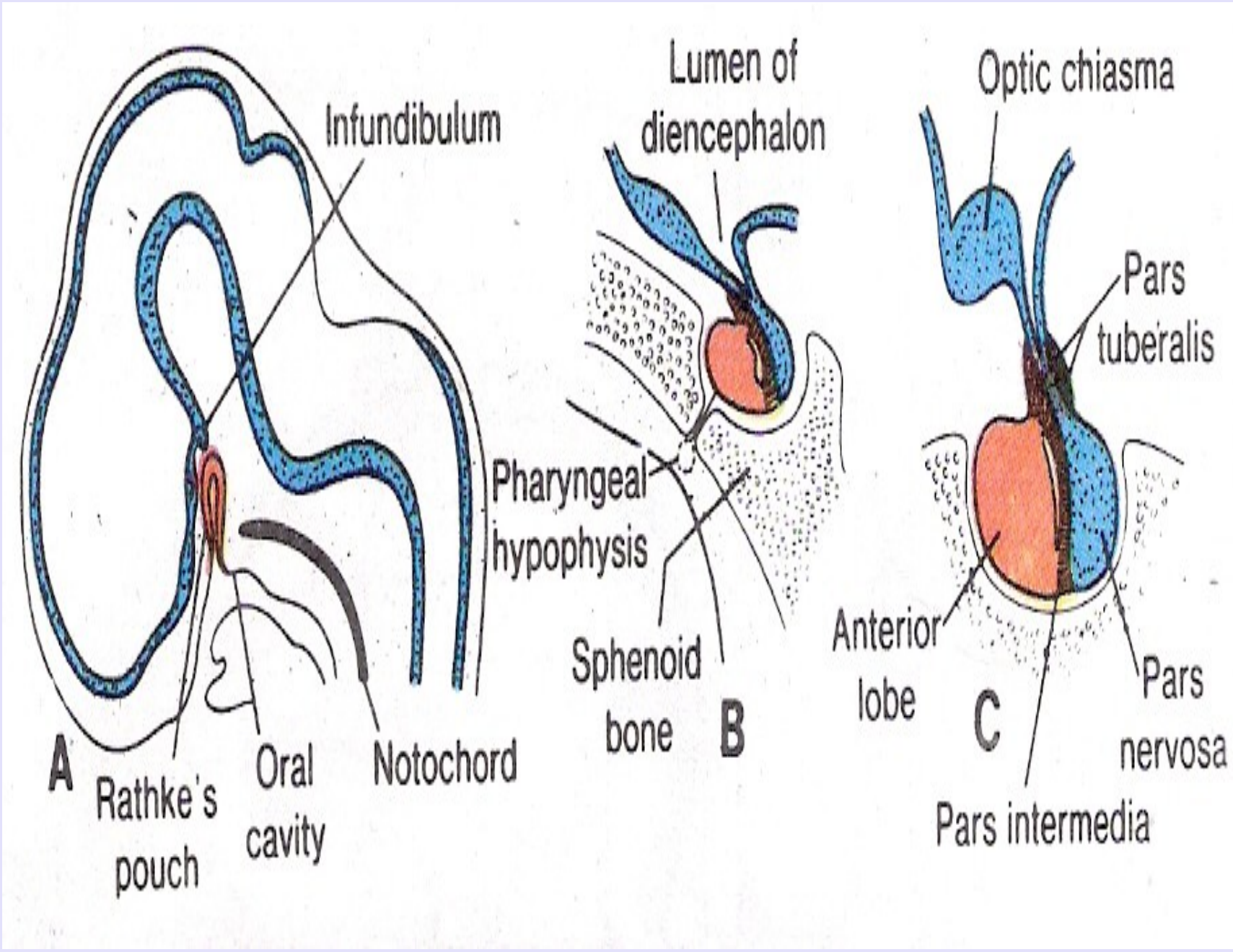
- Or anterior pituitary (glandular part) arise from oral ectoderm.

NEUROHYPOPHSIS:

- Or Posterior Pituitary (Nervous part)
- By the third week hypophyseal diverticulum from Rathke's pouch lies against the ventral wall of forebrain.


Midbrain
Hindbrain
Forebrain
Optic cup
A
Cerebral hemisphere
Cerebellum
Olfactory bulb
Optic nerve
B
Epithalamus
Cerebral hemisphere
Thalamus
Mesencephalon
Alar plate
Sulcus limitans
Basal plate
Cerebellum
Mammillary body
Hypothalamus
Infundibulum
Optic chiasm
C
Pineal gland
Epithalamus
Cerebellum
Thalamus
Hypothalamus
Optic chiasm
Infundibulum
Level of section E
D
Ependymal roof
Epithalamus
Epithalmic sulcus
Third ventricle
Thalamus
Hypothalamic sulcus
Hypothalamus


Hypophysial diverticulum
(upgrowth from roof of stomodeum)
Infundibulum of diencephalon
(downgrowth from floor of forebrain)
Neurohypophysial diverticulum
Floor of diencephalon
Hypophysial diverticulum
Infundibulum
Oral ectoderm
B
C
Notochord
Former site of oropharyngeal membrane
Optic chiasm
Median eminence
Pars intermedia
Pars tuberalis
Anterior lobe
Infundibular stem
Pars tuberalis (anterior lobe)
Pars intermedia
Pars nervosa (posterior lobe)
Developing sphenoid bone
Regressing stalk of hypophysial diverticulum
Colloid-containing vesicles
E
F
Former site of hypophysial stalk
Pharyngeal roof


Infundibulum
Pharyngeal hypophysis
Rathke's pouch
Oral cavity
Notochord
A
Lumen of diencephalon
Sphenoid bone
B
Optic chiasma
Pars tuberalis
Anterior lobe
Pars nervosa
Pars intermedia
C

DEVELOPMENT OF PITUITARY GLAND

IT DEVELOPS FROM TWO DIFFERENT SOURCES:

- An outpocketing of stomodeum
  The Rathke's Pouch forms Adenohyophysis

_A downward extension of the Diencephalon
The Infundibulum forms Neurohypophysis

EMBRYOLOGY OF PITUITARY GLAND

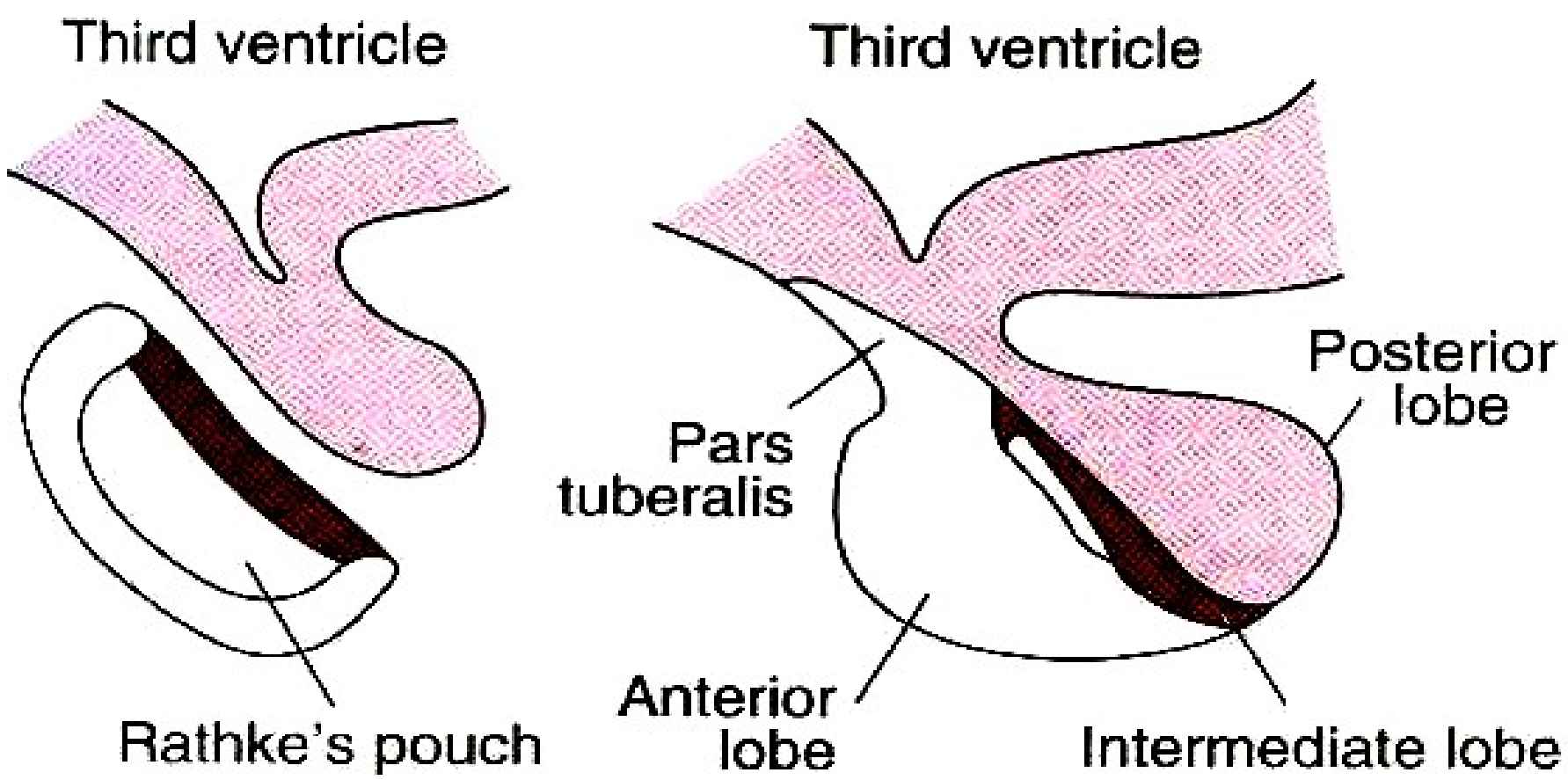


Figure 22–1. Diagrammatic outline of the formation of the pituitary and the various parts of the organ in the adult.

- When embryo is three weeks old Rathke's pouch appears as an evagination of the stomodeum.

  By the end of second month (5th week) it loses its connection with the oral cavity and is then in close contact with the infundibulum.

  The infundibulum give rise to
  -Median Eminence
  - Infundibular stem
  -Pars nervosa

- During further development the cells in the anterior wall of Rathke's pouch increases rapidly and form the anterior lobe of Pituitary gland.

  A small extention of this lobe grows around the stalk of infundibulum The Pars tuberalis.

- The Posterior Wall of Rathke's pouch Gives rise to Pars Intermedia.

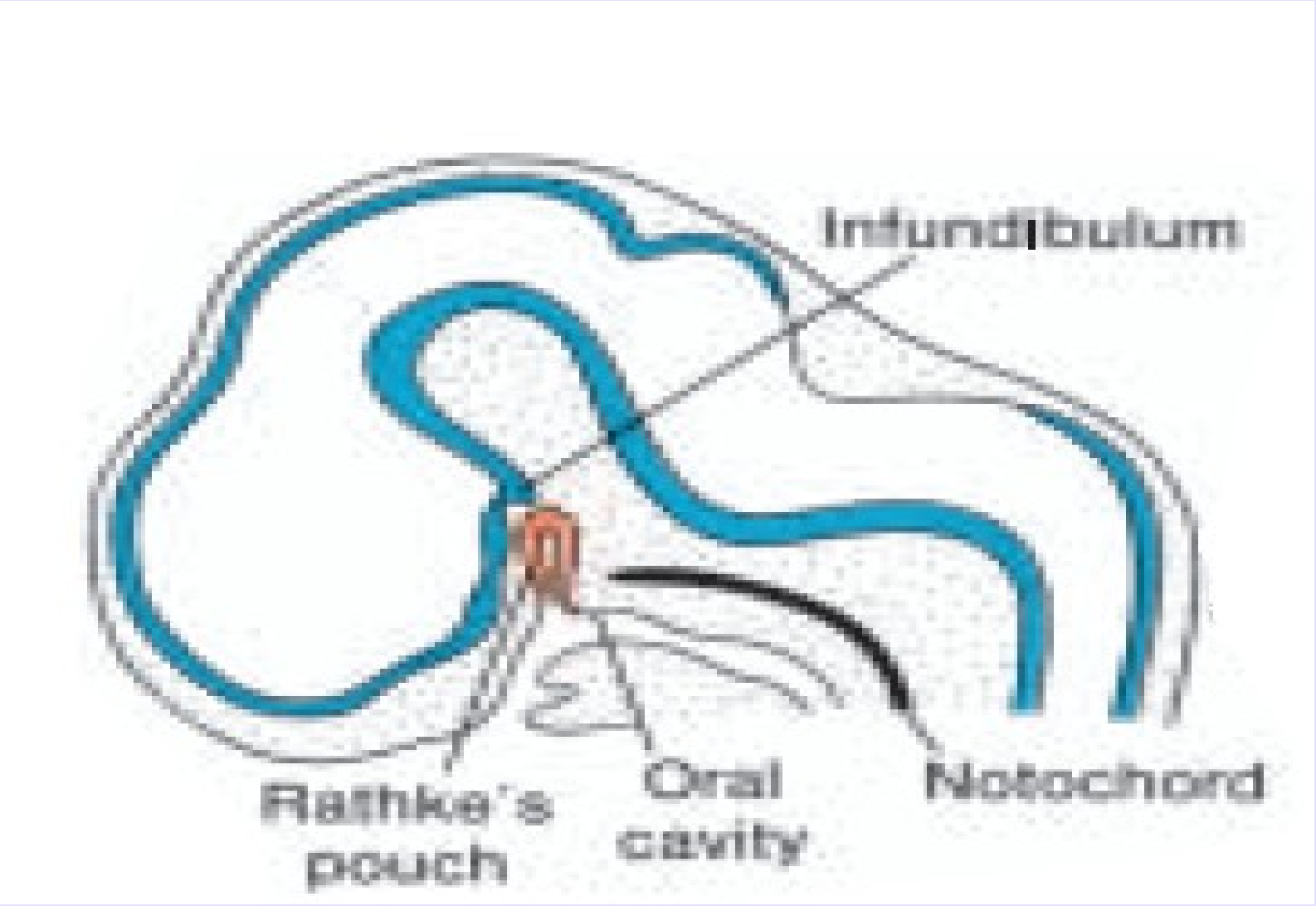
- The Infundibulum Stalk and Pars nervosa of hypophysis contains nerve fibers of neurons located in Hypothalamus & Neuroglia cells.

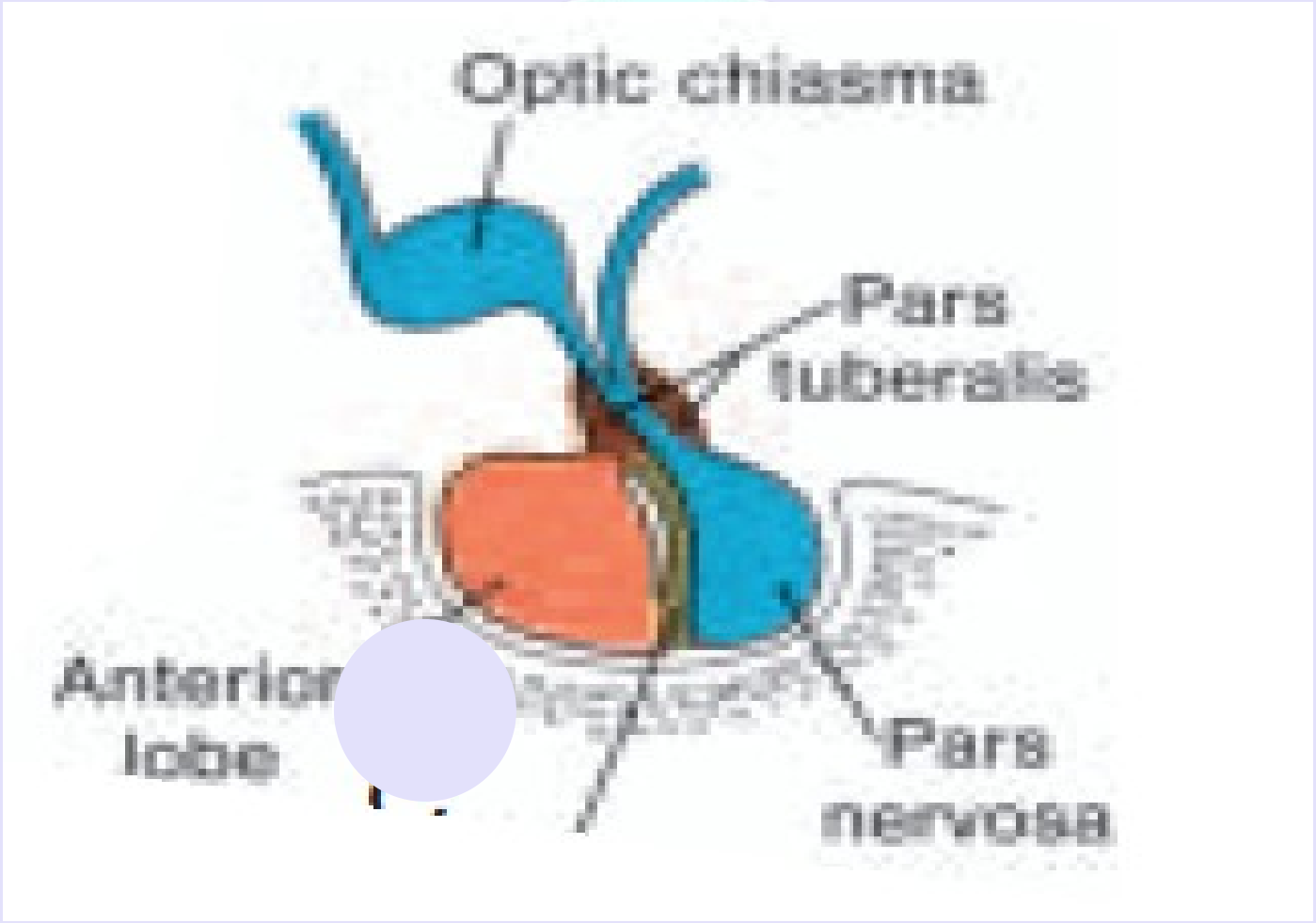
- Sometimes Remanent of Rathke's pouch exist & gives rise Cranipharyngioma.

- The part that develops from ectoderm of stomadeum are pars anterior, pars tuberalis ,pars intermedialis –forms Adenohypophysis.

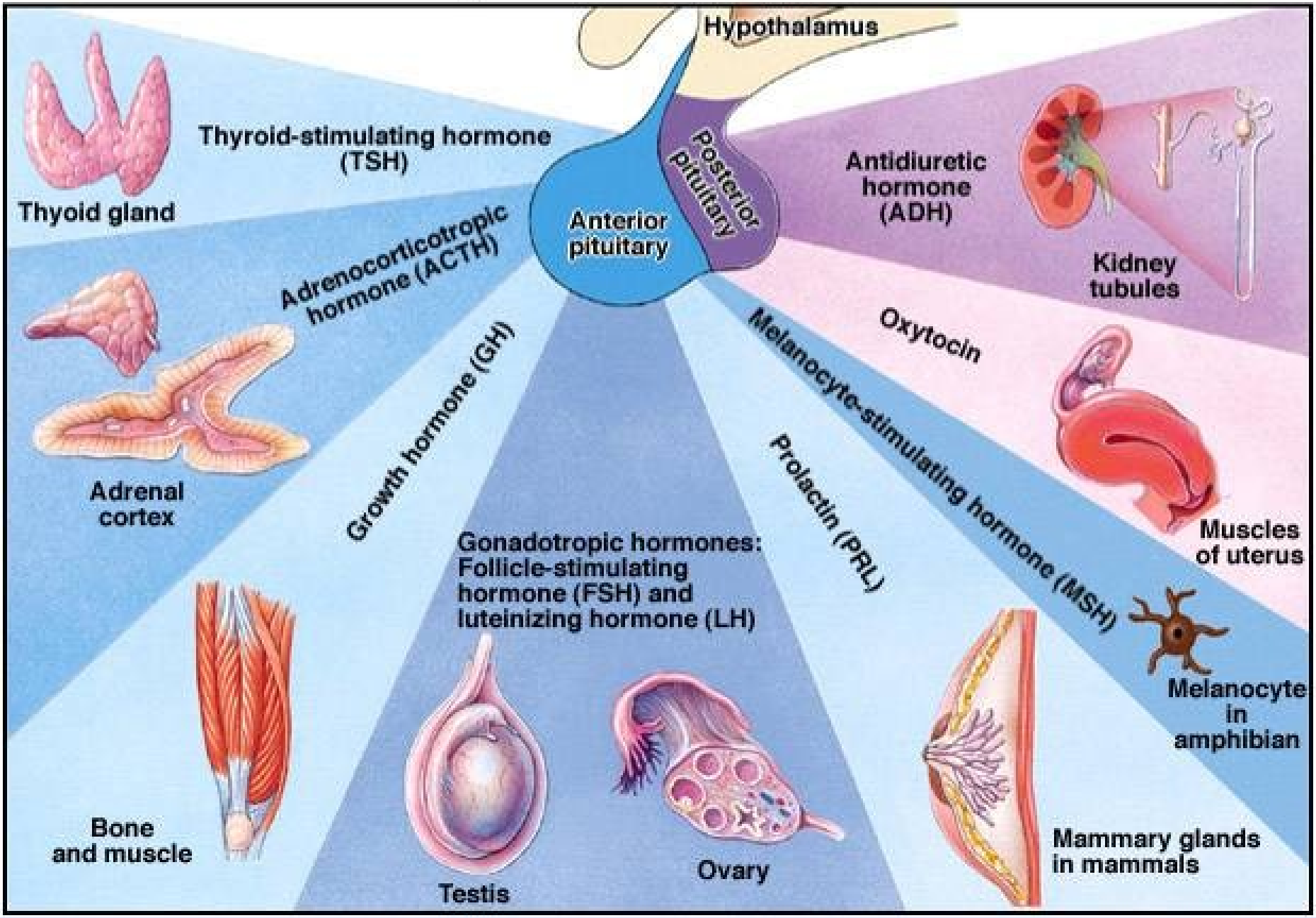
- The stalk becomes compressed during chondrification of Pre sphenoid & basi-sphenoid

- During 6th week the connection of diverticulum with oral cavity degenerates & disappear.


Infundibulum
Rathke's pouch
Oral cavity
Notochord


Optic chiasma
Pars tuberalis
Anterior lobe
Pars nervosa

Pituitary Gland Hormones



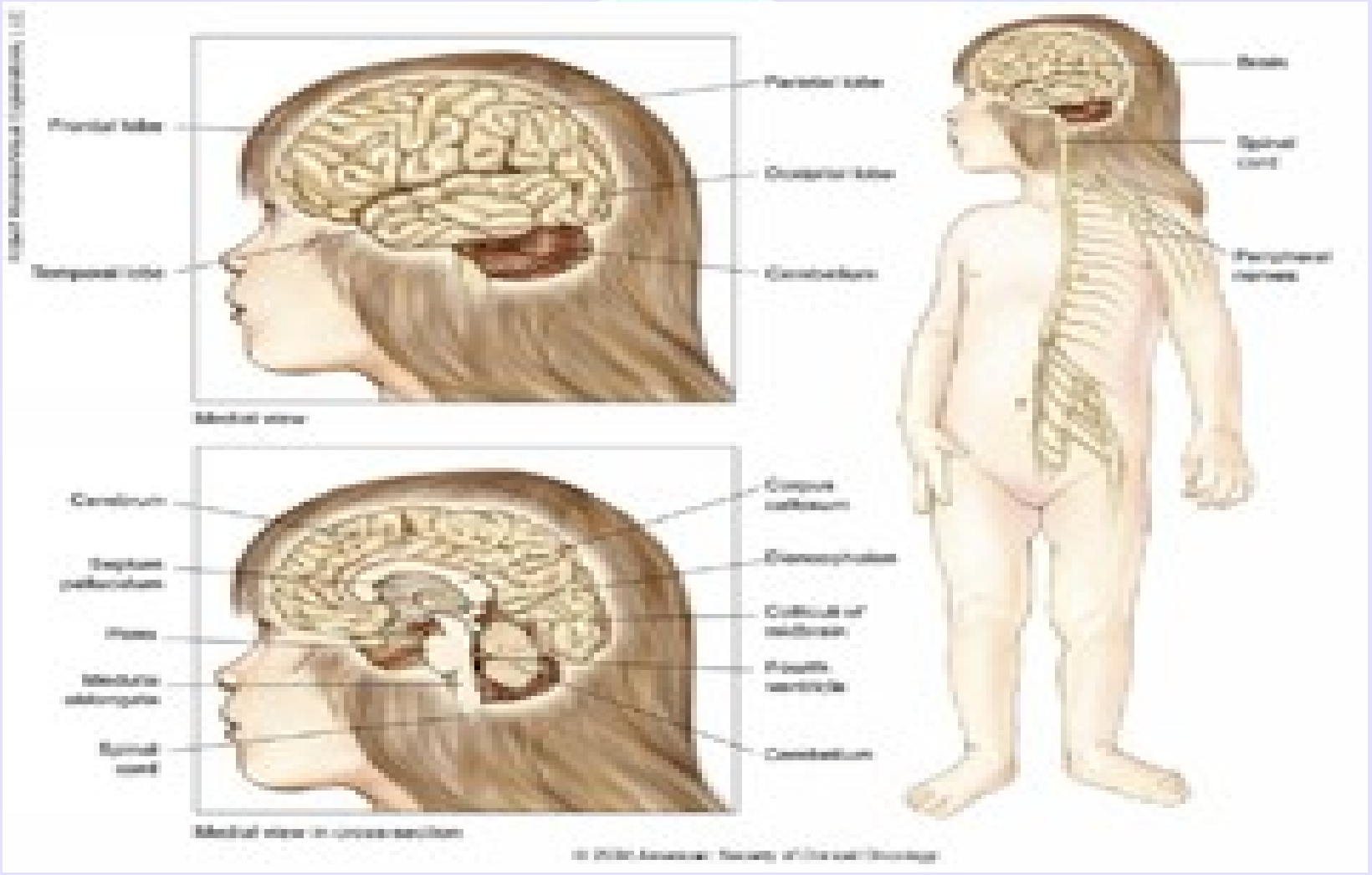
CONGENITAL ANOMALIES

CLINICAL CORRELATES

- A remanent of stalk of Hypophyseal diverticulum may persists & form – **Phayrngeal Hypophysis** in the roof of Oropharynx.

- **Crainophayrngioma-**develops in Phyranx or Basi-sphenoid-remanent of hypophyseal diverticula mostly then form in sphenoid or sella-turcica.

CRANIOPHARYNGIOMA



CONGENITAL ANOMALIES

- A remnant of the hypophyseal diverticulum, the **basipharyngeal canal**, is visible in sections of the newborn sphenoid bone in approx 1% of cases.

REFERENCES

- The Developing Human by K.L.M.

THANK YOU ALL